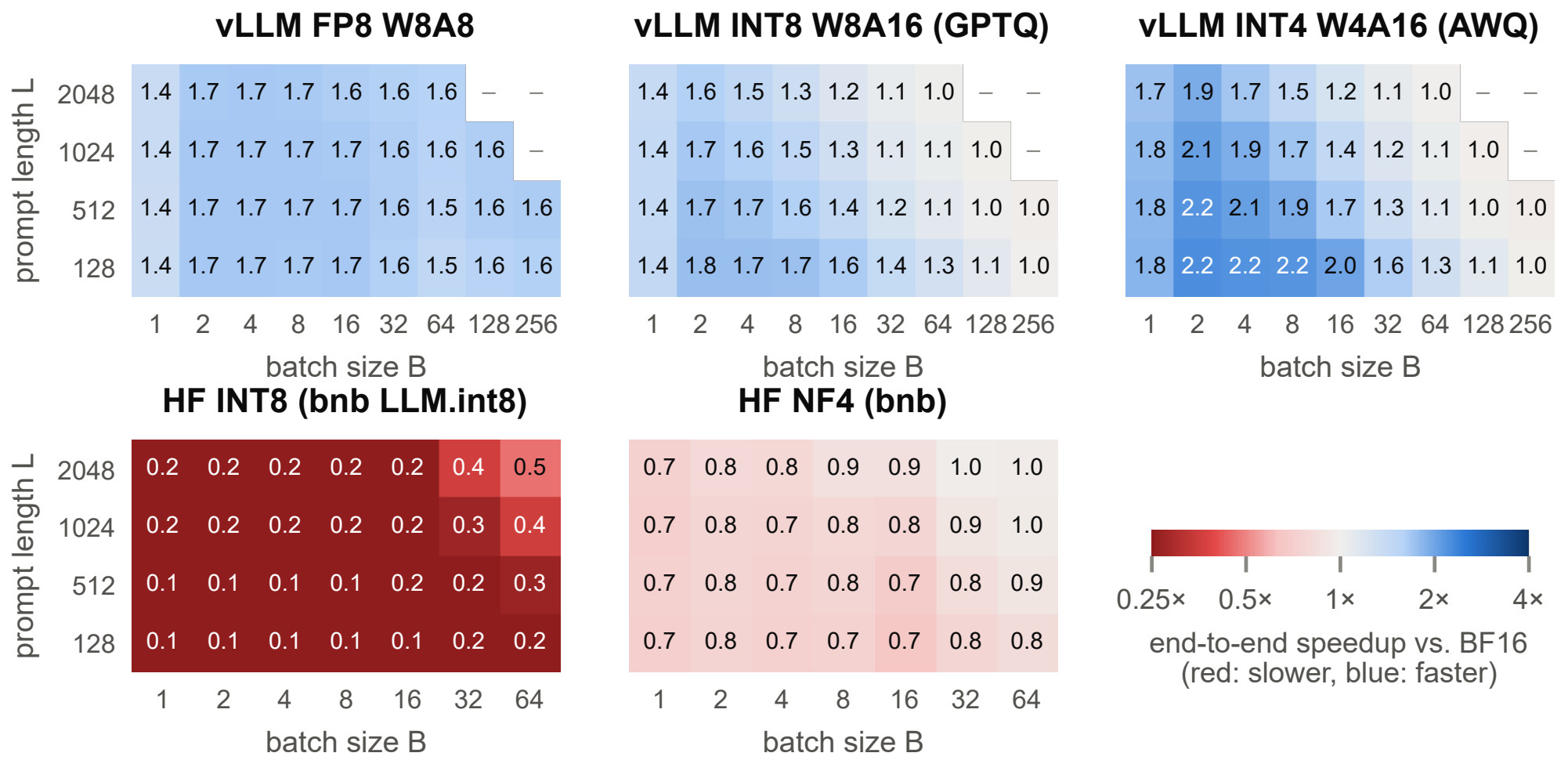
vLLM FP8 W8A8
vLLM INT8 W8A16 (GPTQ)
vLLM INT4 W4A16 (AWQ)
HF INT8 (bnb LLM.int8)
HF NF4 (bnb)
prompt length L
2048
1024
512
128
1 2 4 8 16 32 64 128 256
batch size B
1.4 1.7 1.7 1.7 1.6 1.6 1.6 – –
1.4 1.7 1.7 1.7 1.7 1.6 1.6 1.6 –
1.4 1.7 1.7 1.7 1.7 1.6 1.5 1.6 1.6
1.4 1.7 1.7 1.7 1.7 1.6 1.5 1.6 1.6
1.4 1.6 1.5 1.3 1.2 1.1 1.0 – –
1.4 1.7 1.6 1.5 1.3 1.1 1.1 1.0 –
1.4 1.7 1.7 1.6 1.4 1.2 1.1 1.0 1.0
1.4 1.8 1.7 1.7 1.6 1.4 1.3 1.1 1.0
1.7 1.9 1.7 1.5 1.2 1.1 1.0 – –
1.8 2.1 1.9 1.7 1.4 1.2 1.1 1.0 –
1.8 2.2 2.1 1.9 1.7 1.3 1.1 1.0 1.0
1.8 2.2 2.2 2.2 2.0 1.6 1.3 1.1 1.0
0.2 0.2 0.2 0.2 0.2 0.4 0.5
0.2 0.2 0.2 0.2 0.2 0.3 0.4
0.1 0.1 0.1 0.1 0.2 0.2 0.3
0.1 0.1 0.1 0.1 0.1 0.2 0.2
0.7 0.8 0.8 0.9 0.9 1.0 1.0
0.7 0.8 0.7 0.8 0.8 0.9 1.0
0.7 0.8 0.7 0.8 0.7 0.8 0.9
0.7 0.8 0.7 0.7 0.7 0.8 0.8
0.25× 0.5× 1× 2× 4×
end-to-end speedup vs. BF16
(red: slower, blue: faster)